GRADE 10 GENERAL SCIENCE - PHYSICS LABORATORY MANUAL

NexSchool Enterprise Education Platform | Academic Session 2025-2026

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1. Resource Overview & Safety Protocols

Subject: General Science (Physics Mechanics Module) | Resource Type: Syllabus Lab Manual
Published by: Sarah Jenkins, Senior Academic Faculty | Academic Session: 2025-2026
Syllabus Coverage: Dynamics, Kinematics, Friction, and Free Body Force Analysis

2. Experiment 1: Verification of Newton Second Law (F = ma)

Apparatus: Low-friction air track, glider, hanging weight carrier, photogates, digital timer.
Theory: Newton second law states that net force equals mass multiplied by acceleration.
Procedure: Vary hanging mass m_h while keeping total system mass constant. Measure acceleration using dual infrared photogates across 0.50m and 1.00m distance markers.
Analysis: Plot Net Force vs Acceleration. The slope represents experimental system mass.

3. Experiment 2: Measurement of Friction Coefficients on an Inclined Plane

Apparatus: Adjustable wooden inclined track, protractor, friction block, spring balance.
Theory: Static friction coefficient mu_s = tan(theta_c), where theta_c is the angle of impending motion.
Calculations: Record 5 trials of critical slipping angle. Compute mean angle and standard error.

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